Artificial Intelligence Driven Meta-Analysis

An Overview of Evidence-based Medicine

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Faculty of Applied Sciences

Macao Polytechnic University

Curriculum Vitae



Academic background

- Lecturer, Faculty of Applied Sciences, Macao Polytechnic University
- Ph.D., University of Macau | Postdoc, McGill University
- Master's & Ph.D. Supervisor

Web of Science Core Collection metrics

24	82
H-Index	Publications

Academic performance

- Research Focus: AI + Mental Health
- Editorial Board Members: *Alpha Psychiatry, Plos digital health, etc.*
- Peer Reviewers: *JAMA Network Open, JMIR, BMJ Public Health, Plos digital health,* etc.
- Published 80^{+} papers, Web of Science H-index: 24

FACULTY OF APPLIED SCIENCES

PHD OF ARTIFICIAL INTELLIGENCE EMPOWERED SMART HEALTHY AGEING

LEARNING MODULE OUTLINE

Academic Year	2026/2027	Semester	1
Module Code	AIHA8123		
Learning Module	Artificial Intelligence Driven Meta-Analysis		
Pre-requisite(s)	Nil		
Medium of Instruction	English		
Credits	3	Contact Hours	45 hrs
Instructor	Dr. Wenwang Rao	Email	raoww@mpu.edu.mo
Office	ACADEMIC BUILDING; E1303-2	Office Phone	8599 3826

Times	Assessment Activities	Weighting (%)
2	A1. Assignment / Classwork	25
1	A2. Tests	25
1	A3. Project	50



In-class assignments



The course module outline, PowerPoint slides, assignments, and other materials will be uploaded to Canvas. If you have any questions, please email raoww@mpu.edu.mo.

My office hours are at 10.00 am to 1.00 pm on Tuesday and on Thursday.

My office is currently located at the Academic Building F1303-2 (教研樓) or Room 206, Building 11, Dezhi Plaza, Macao-Hengqin International Education (University) Town.

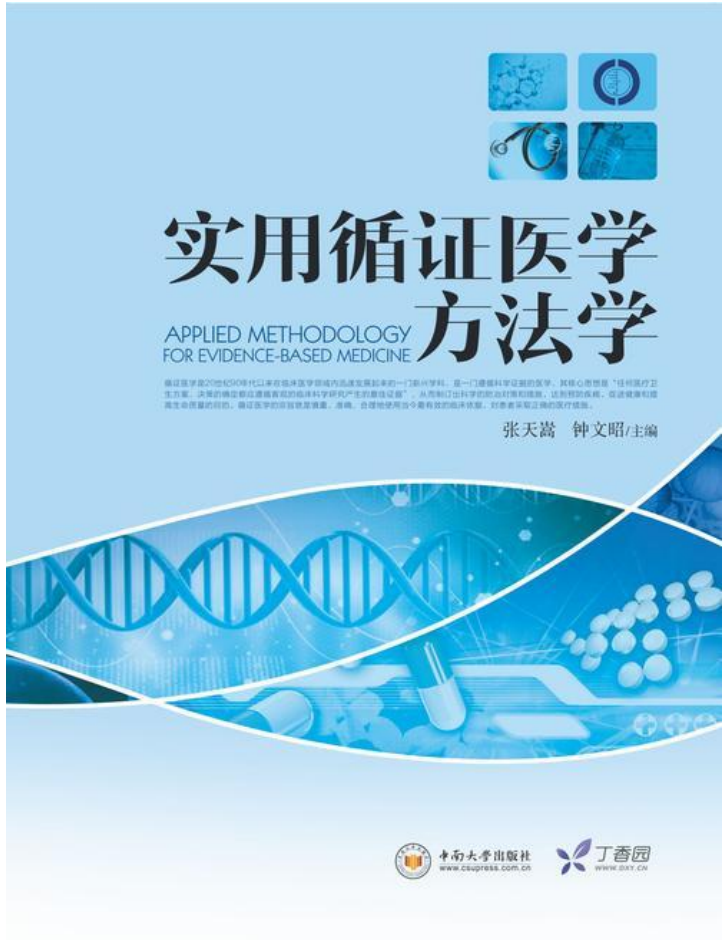
Attendance

- Attendance requirements are governed by AIHA of the Macao Polytechnic University.
- Students are required to maintain a minimum attendance rate of 70%.
- Students who do not meet the attendance requirements for the learning module shall be awarded an 'F' grade.

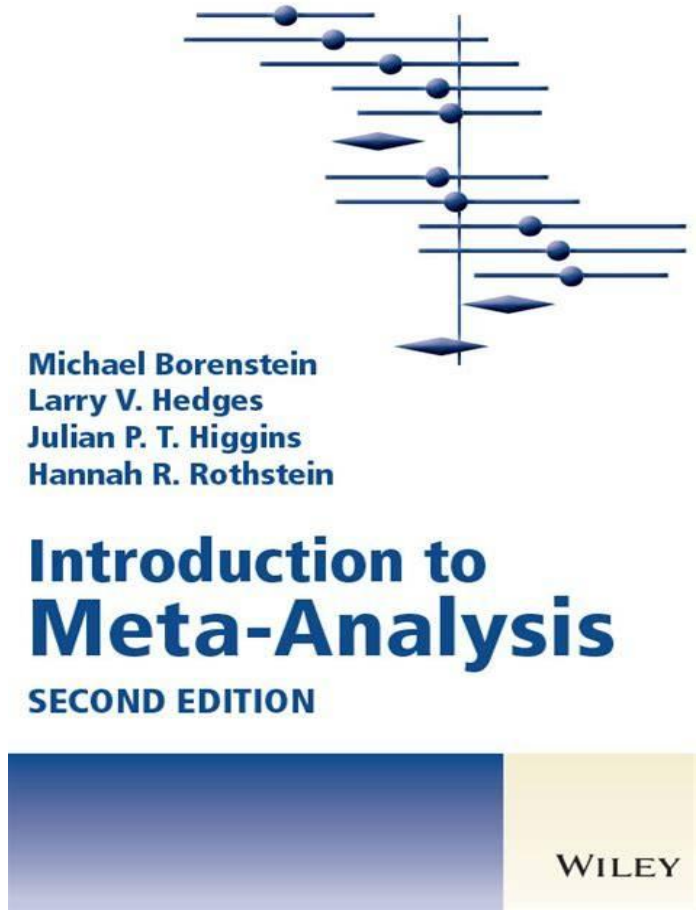
  - For Chinese students, full Chinese/English names are required;
  - For Foreign students, full names or nicknames are acceptable;
  - Attendance are based on the sign-in sheet only.



Course Reference Book



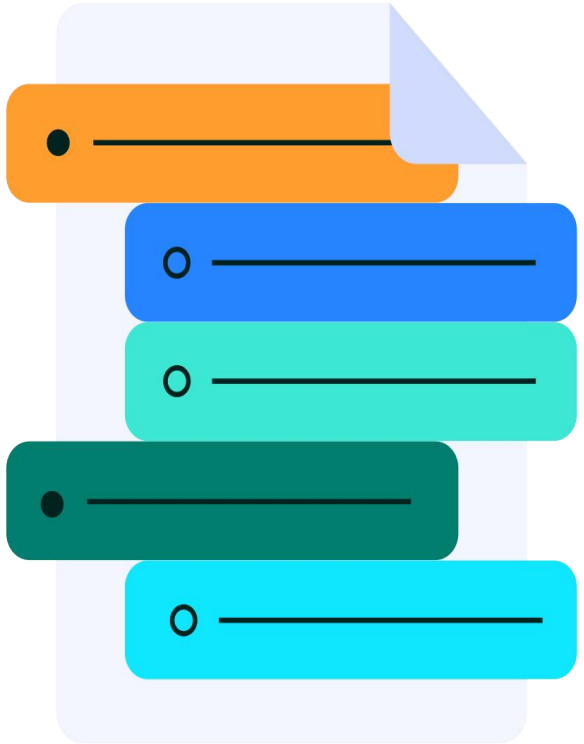
Chinese Version



English Version

Overview

- Introduction to EBM
- Define EBM
- Learn steps in EBM process
- Identify parts of a well-built clinical question
- Discuss resources for literature search
- Critical appraisal of the evidence
- Apply to the patient
- Clinical applications



Bloodletting



- Bloodletting originated approximately 3000 years ago
- Adopted by the ancient Egyptians, Greeks then Romans, Arabs and so on.
- The cure for (hot, moist diseases) several medical conditions.

Pierre Louis (1787-1872)

Inventor of the "numeric method" and the "method of observation"

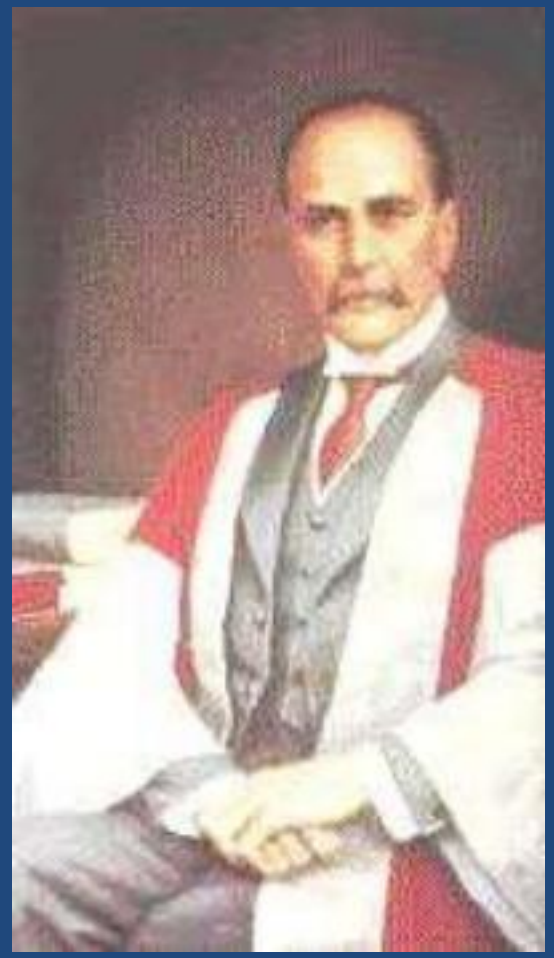


- French physician who wanted to analyze the efficacy of bloodletting in the treatment of acute pneumonia
- Examined the clinical course and outcomes of 77 patients
- Conclusion: Effect of bloodletting procedure was actually much less helpful than has been commonly believed

Early pioneer of EBM

William Osler (1849 -1919)

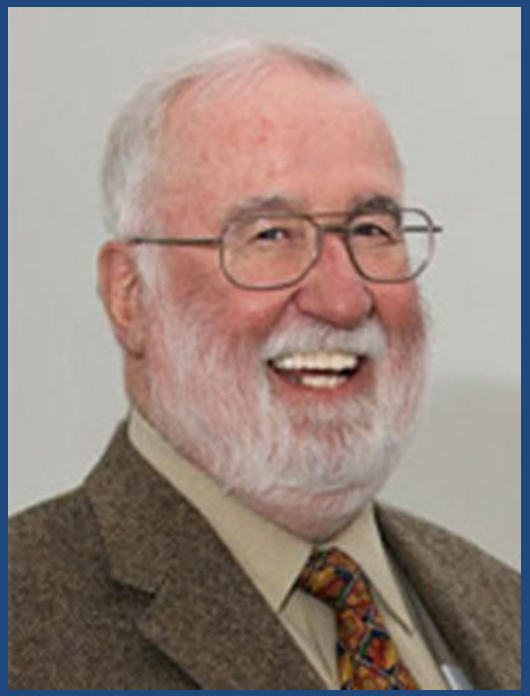
First “attending physician” at Johns Hopkins



- Author of hugely influential textbook, ‘The Principles and Practice of Medicine’
- believed that most drugs in his day were useless, ***but still advocated blood-letting*** in some cases

Definition of EBM

- The conscientious, explicit, judicious use of current best evidence in making decisions about the care of individual patient.
- It means integrating individual clinical expertise with the best available external clinical evidence from systematic research.



Dr. David Sackett, 1996

Father of evidence-based medicine (EBM)

UIC LIBRARY OF THE HEALTH SCIENCES

Introduction to Evidence Based Medicine (EBM)

The Evidence-Based Medicine triad

(see D.L. Sackett et al, BMJ 1996; 312: 71-72)



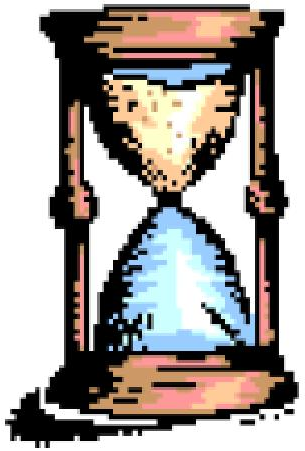
Why is EBM important?

- New types of evidence are being generated which can create changes in the way patients are treated

How much is actually being applied to patient care?

- Although evidence is needed on a daily basis, usually physicians don't get it:
  - lack of time
  - out-of-date textbooks
  - the disorganization of the up-to-date journals

Importance of EBM for practicing clinicians?





TIME AVAILABLE TO READ:

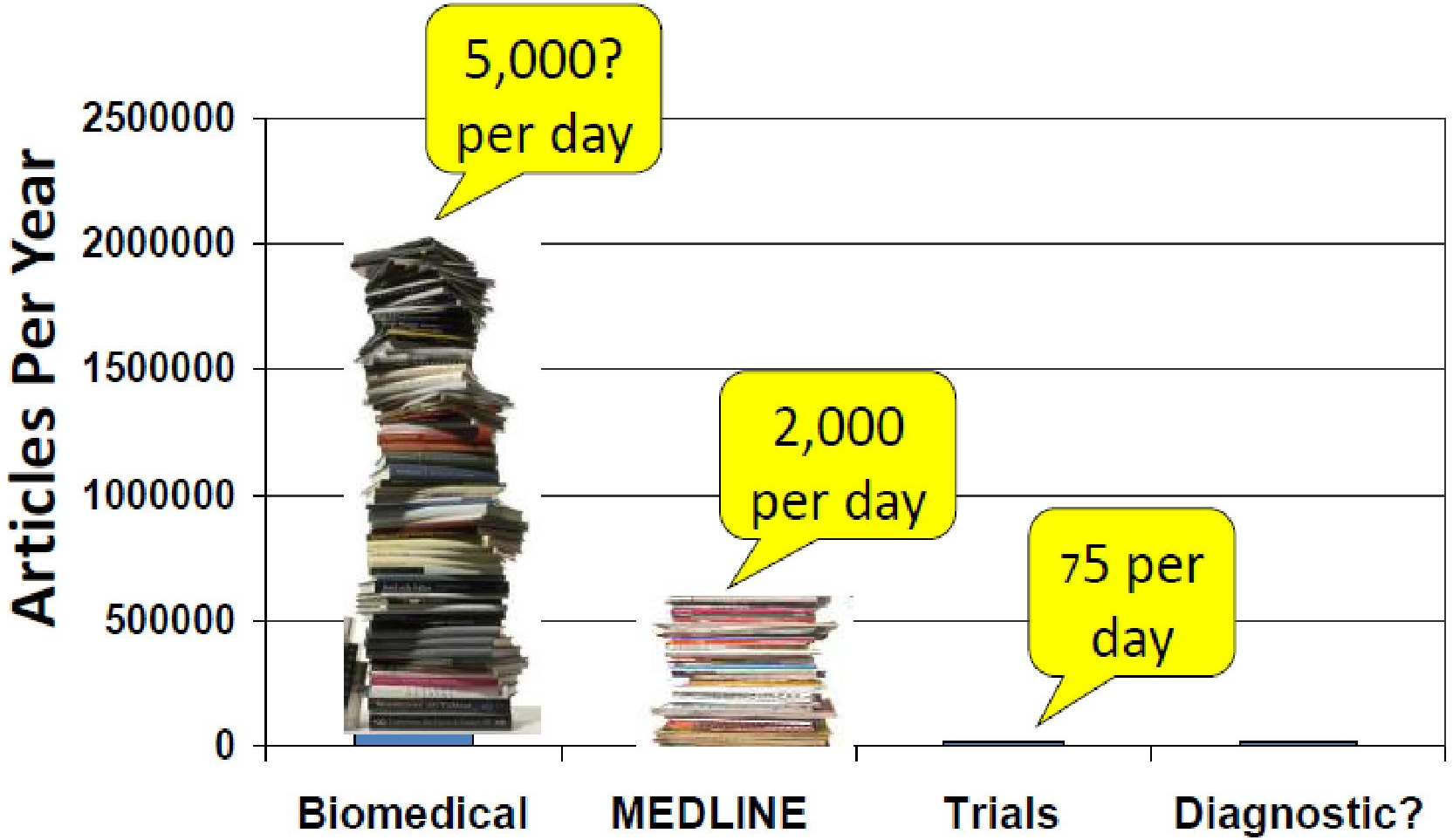
Less than
1 Hour per Week

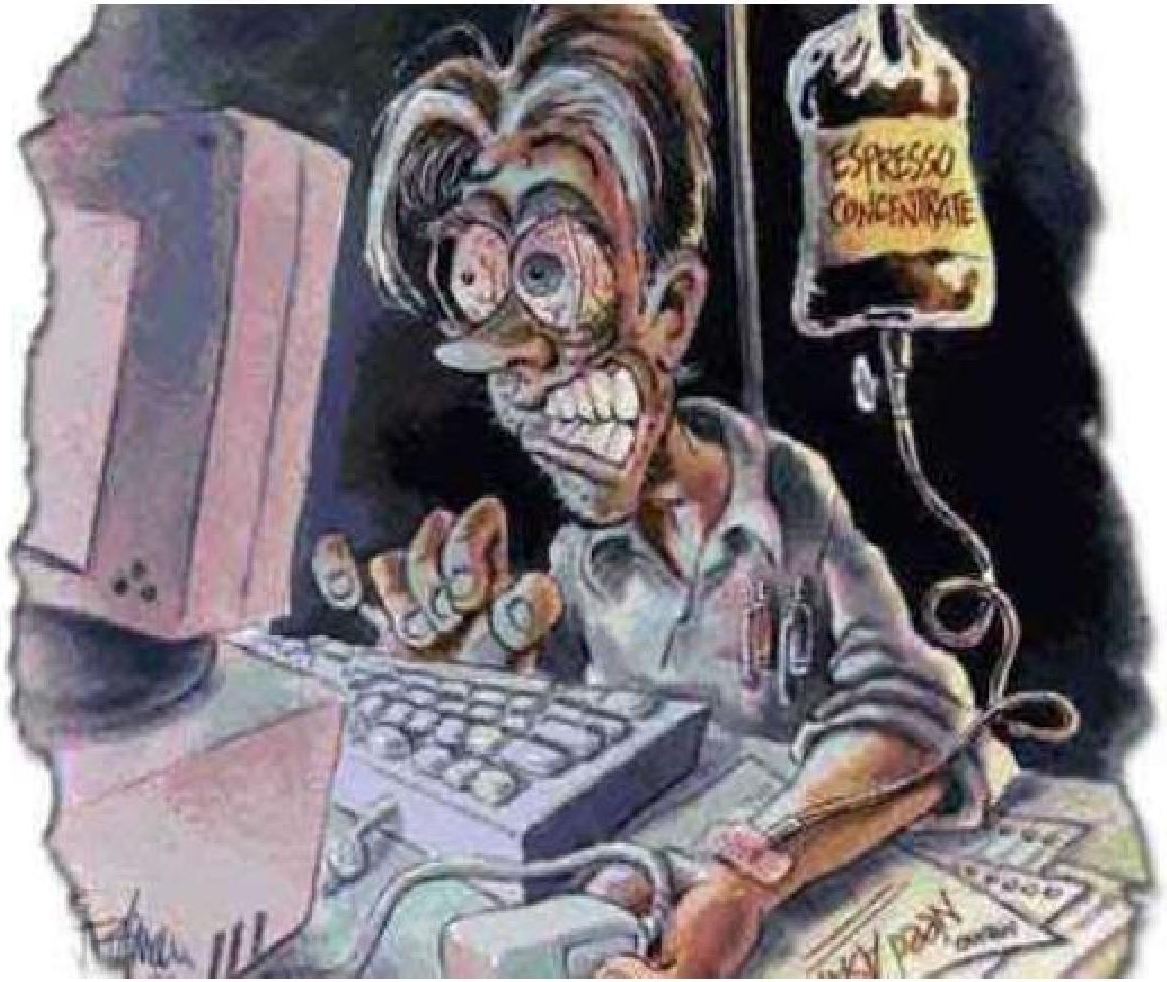
TIME NEEDED TO KEEP CURRENT ON GENERAL MEDICINE:

19 Articles per DAY
365 Days per Year

Source: Davidoff F, Haynes B, Sackett D, Smith R. *BMJ.* 1995;310:1085-1086.

Why do we need to use evidence efficiently?

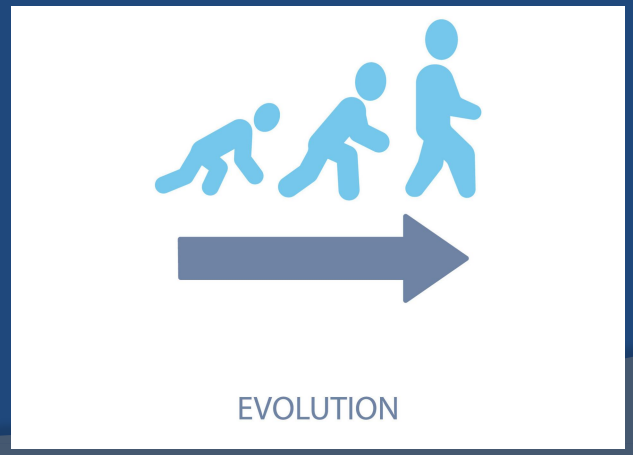




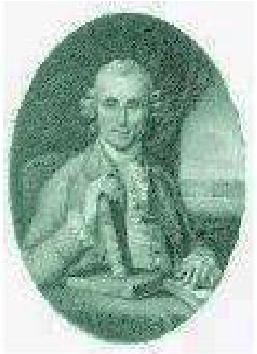
Time-poor clinician suffering from Information Overload

Evolution of EBM

- **Pre EBM:** Passive diffusion (“publish it and they will come”)
- **Early EBM:** Pull diffusion (“teach them to read it and they will come”)
- **Current EBM:** Push diffusion (“read it for them and send it to them”)
- **Future EBM:** Prompt diffusion (“read it for them, connect it to their individual patients”)



Some milestones in the history of EBM



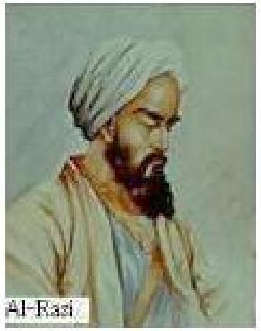
James Lind
publishes review & clinical trial in *Treatise on Scurvy*



Bradford-Hill
publishes *Principles of Medical Statistics* &
MRC trial of streptomycin

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900 AD | 1780 | 1840 | 1937/48 | 1967 | 1970's



Al-Rhazi

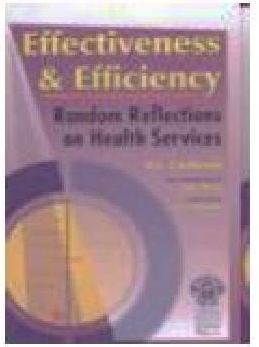
For I once saved one group by it, while I intentionally neglected another group. By doing that, I wished to reach a conclusion.



Pierre Louis
Develops his "numerical method" and changes blood letting practice in France



Alvan Feinstein
publishes his book *Clinical Judgement*

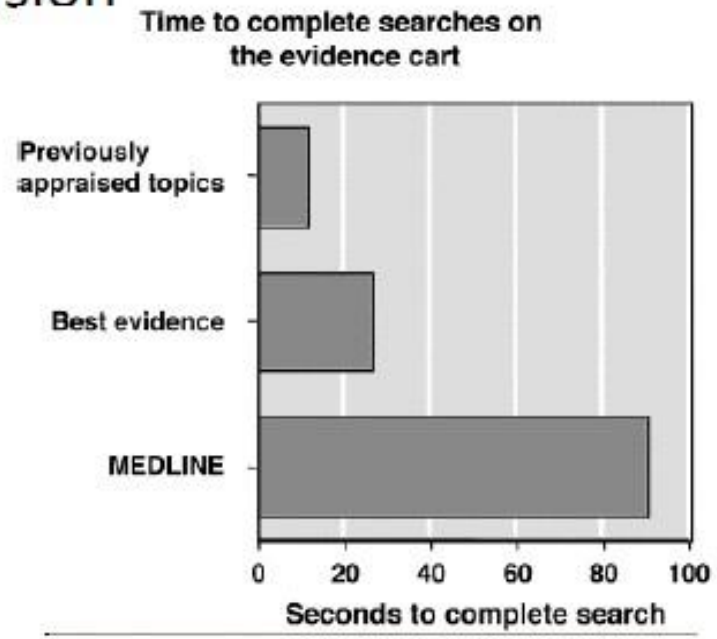


An EBM Approach to Education

- Evidence cart on ward rounds - 1995
- Looked up 2-3 questions per patient
- Took 15-90 seconds to find
- Change about 1/3 decision

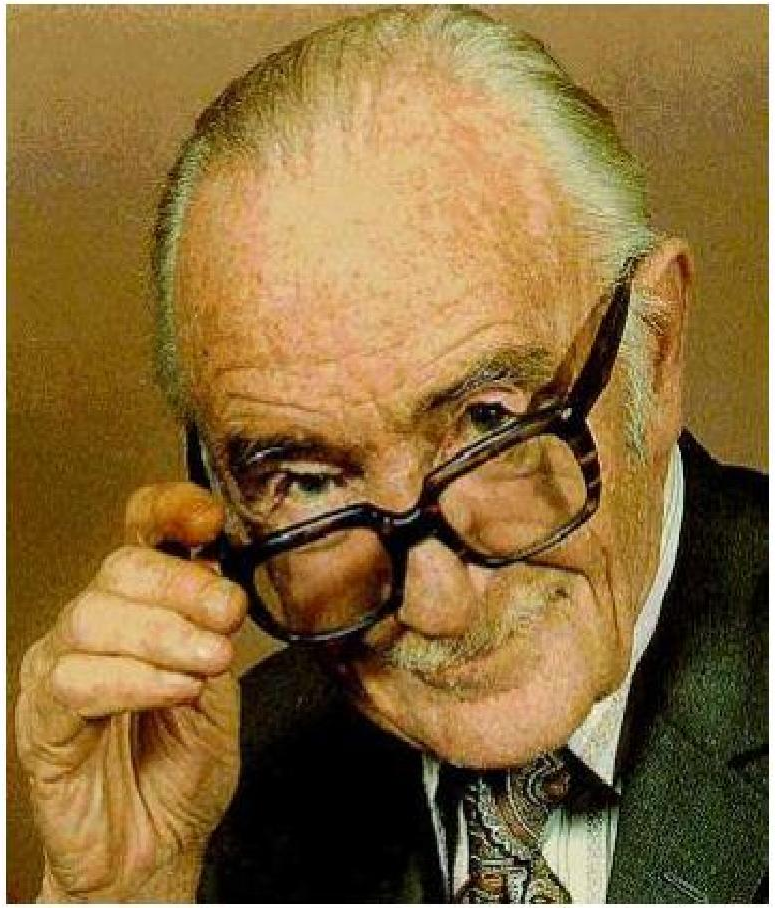


David Sackett



Prof Archibald Cochrane, CBE
(1909 - 1988)

- The Cochrane Collaboration is named in honor of Archie Cochrane, a British researcher.
- In 1979 he wrote, **"It is surely a great criticism of our profession that we have not organized a critical summary, by specialty or subspecialty, adapted periodically, of all relevant randomized controlled trials"**



When…

- There is evidence that something works, is good and benefits the patient- *do it*
- There is evidence that something does not work, is harmful or does not benefit the patient- *do not do it*
- There is insufficient evidence, rely on individual clinical expertise- *be conservative*

Misconceptions about EBM

×× It ignores clinical experience and clinical intuition.

×× Understanding of basic investigation and pathophysiology plays no part in it.

×× It ignores standard aspects of clinical training such as the history taking, physical examination etc.

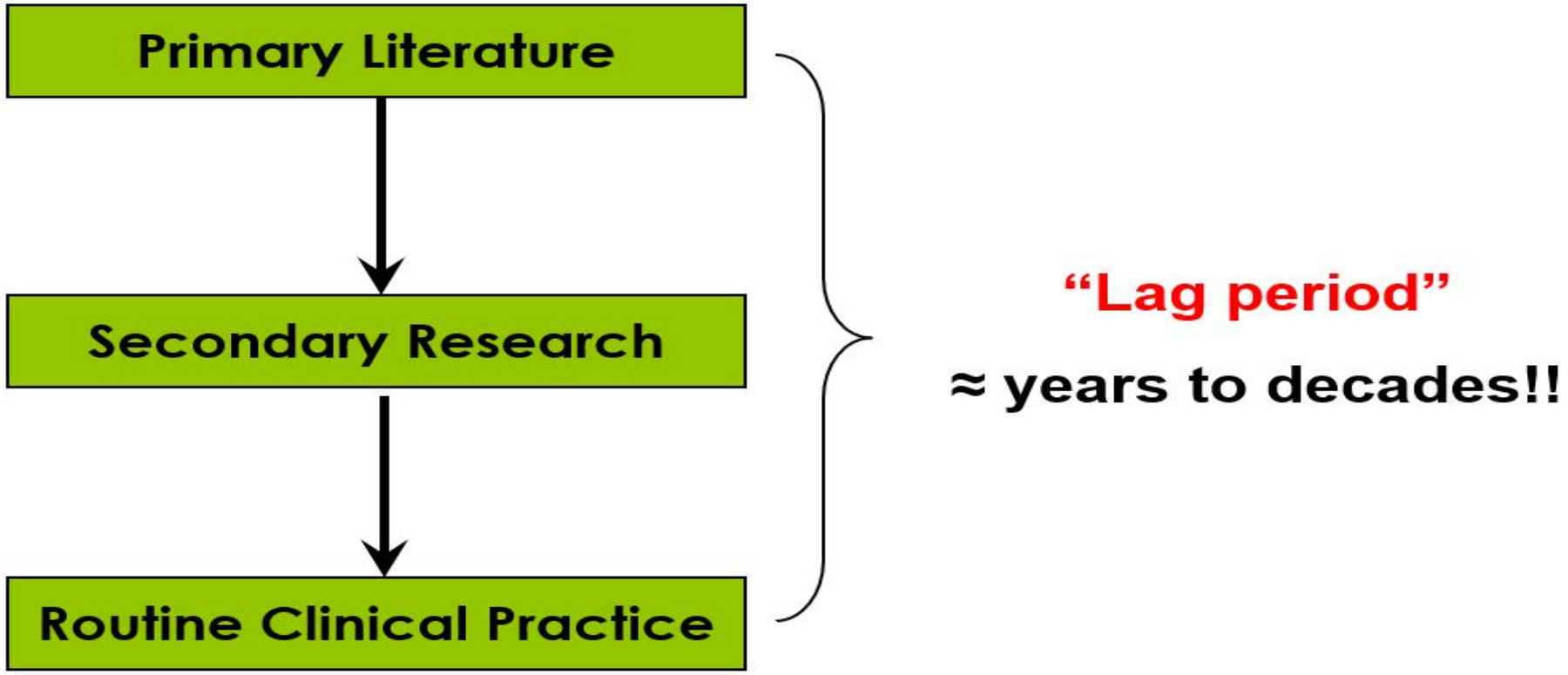
Why EBM?

- Cost
- Delay of "bench-to-bedside" research
- Managing the literature
- Counter misleading marketing
- Dealing with conflicting results

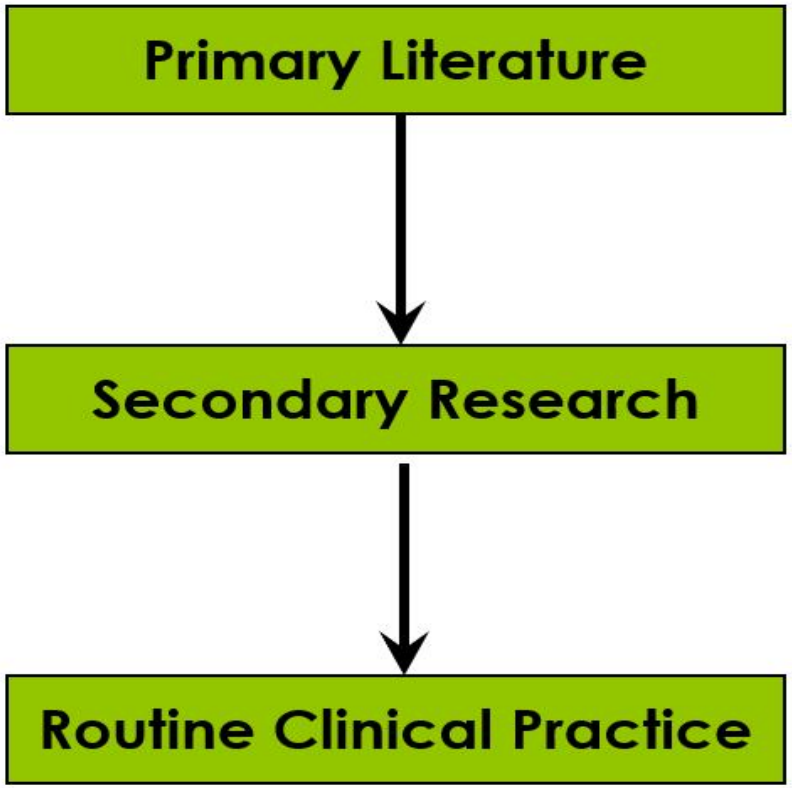
Cost

- Many companies often use cost-cutting measures (such as treatment algorithms) under the name of EBM.
- Increasing pressure to demonstrate effectiveness of interventions.
- When cost is a barrier for a patient, it is important for clinicians to know when treatments are wholly ineffective and make decisions to utilize the most cost effective measures.

Delay of "bench-to-bedside" research



Delay of "bench-to-bedside" research



Thrombolytics and Aspirin for acute MI: 6 years from the first Systematic Reviews of RCTs until most review articles, textbooks and expert opinions recommended their use.

(Antman EM, Lau J, Kupelnick B, Mosteller F, Chalmers TC. A comparison of results of meta-analyses of randomized control trials and recommendations of clinical experts. Treatments for myocardial infarction. JAMA 1992; 268(2): 240-8.)

Managing the literature

- 60,000 articles/year from 120 reputed journals worldwide.
- More than 3800 biomedical journals in MEDLINE, more than 7300 citations added weekly
- Just within their own fields, physicians would need to read 19 articles per day, 365 days per year, to keep up with research.
- Not all (~10%) of these articles are considered high quality and clinically relevant.

Thus EBM helps us to find the most appropriate article for a specific clinical question

Counter misleading marketing

- **Pharmaceutical companies invest considerable resources to promote products based on skewed or selective evidence or through direct-to-consumer advertising.**
- **EBM provides tools to alert clinicians against potentially misleading marketing.**
- **One such tool to detect publication bias is *clinical trial registries*, which also guard against data mining by "post hoc" statistical analysis.**

Dealing with conflicting results

A counter-intuitive result, demonstrating the inability to make accurate predictions based on physiologic reasoning, theoretical knowledge or results of studies which are biased or having poor methodological flaws.

Some examples:

- **Beta-blockers were initially avoided after MI thinking that they would decrease compensatory sympathetic mechanisms.**
- **Later shown to decrease hospitalization & death.**

Principles of EBM

① Construct a well-built clinical question and classify it into one category (therapy, diagnosis, etiology, prognosis, prevention or cost)

② Find the evidence in health care literature

③ Critically appraise or formally evaluate for validity and usefulness

④ Integrate the evidence with patient factors

⑤ Evaluate the whole process

Why the sudden interest in EBM?

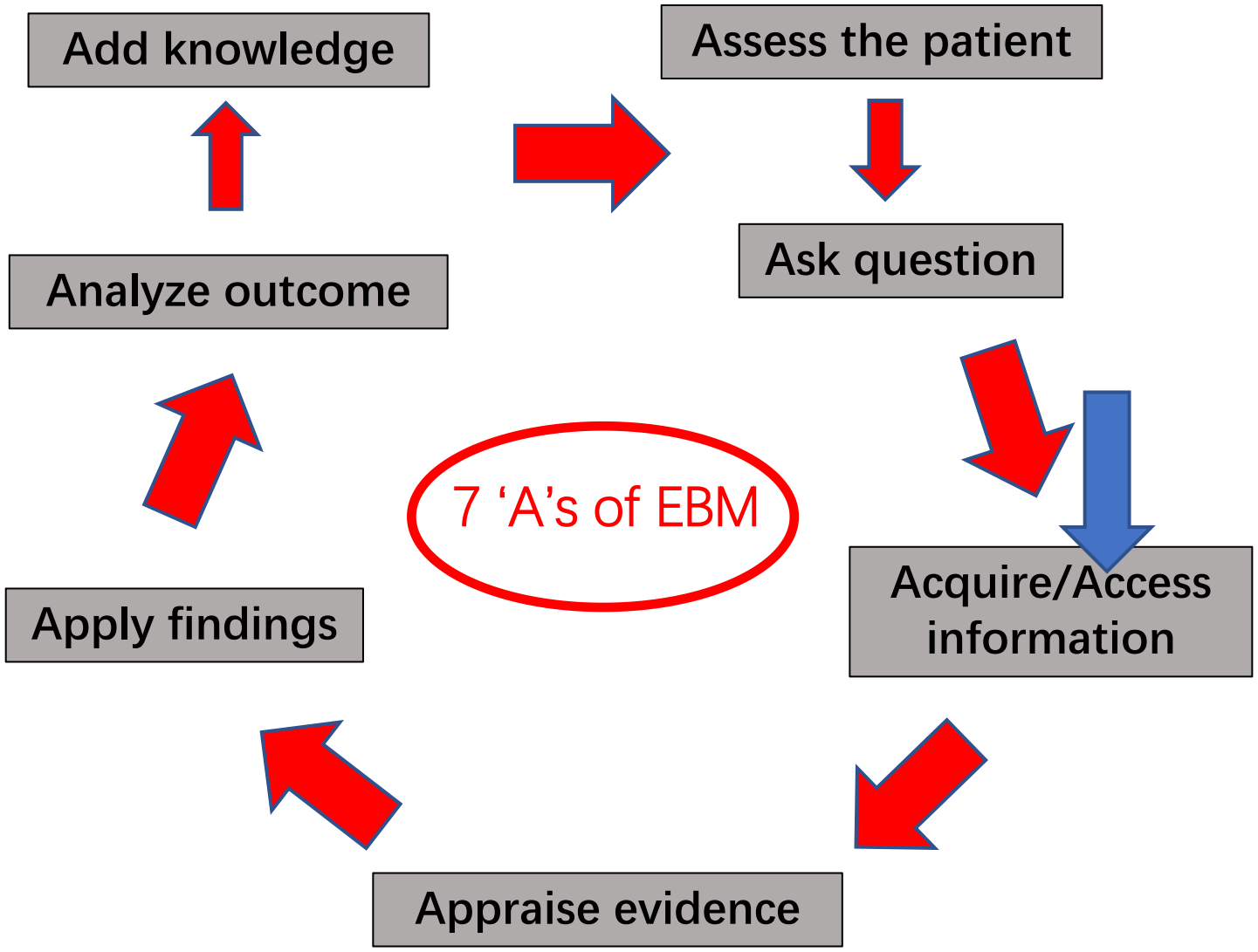
- Increasing realization among clinicians that years of experience unaccompanied by updating of knowledge can result in decline of clinical performance
- The need for valid information about diagnosis, therapy, prognosis, and prevention in this era of consumer activism



- **Limited time available to the clinician for acquiring information is a major impediment for updating the knowledge from traditional sources**

5 A's - Steps in EBM Process

- Assess the patient – a clinical problem or question arises from care of the patient
- Ask the question – construct a well-built clinical question
- Acquire the evidence – select the appropriate resources and conduct a search
- Appraise the evidence – check for validity and applicability
- Apply the evidence – integrate with clinical expertise and patient preferences and apply it to practice



- Analyze outcomes checks whether the evidence-based action really worked in practice.
  - Did it work?
- Add knowledge transforms clinical experience and outcomes into reusable knowledge for future decisions.
  - What can we learn from it and use next time?

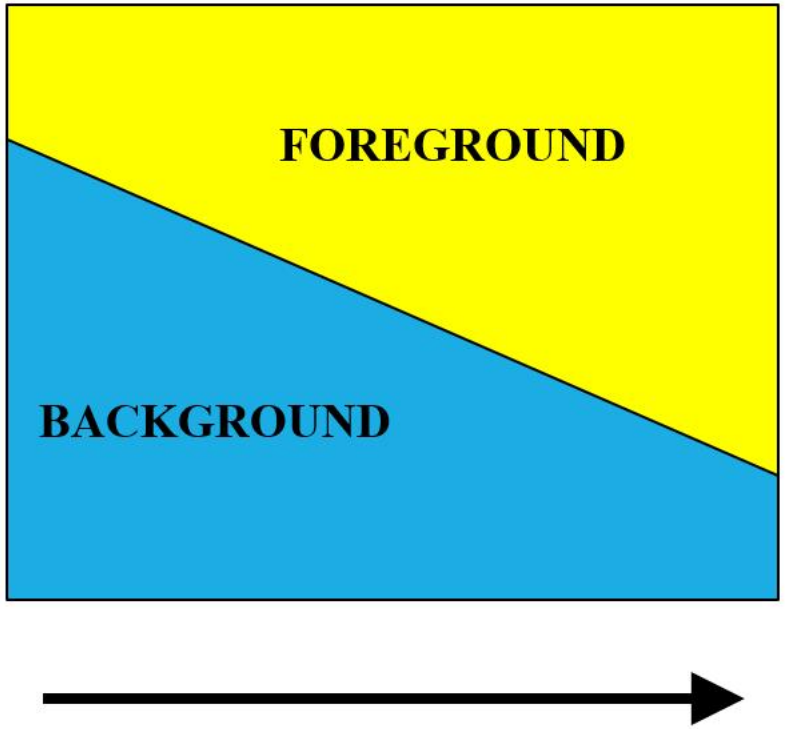
Background vs. Foreground Questions

- Background questions
  - Very general
  - Apply to most patients
  - Basic aspect of a disease
    - pathophysiology
    - etiology
    - basic treatment
  - Who, what, when, how

Background vs. Foreground Questions

- Foreground questions
  - Relate to specific aspects of a given patient
  - Specific knowledge
  - 4 parts
    - Patient/problem
    - Intervention
    - Comparison
    - Outcomes

Type of Question



- One can think of **first set of questions**, those of **medical student**, as **background questions** and of **browsing and problem-solving sets** as **foreground questions**
  - In most situations→ you need to understand background thoroughly before it makes sense to address foreground issues

Background vs. Foreground

- **Identify the following questions as either background or foreground questions:**

1. **What causes gastroenteritis?**
2. **Is oral rehydration as effective as IV rehydration?**
3. **How can I tell if my patient is dehydrated?**


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Scan QR Code

Background vs. Foreground

- **Identify the following questions as either background or foreground questions:**

1. **What causes gastroenteritis?**

   Background

2. **Is oral rehydration as effective as IV rehydration?**

   Foreground

3. **How can I tell if my patient is dehydrated?**

   Background

Background vs. Foreground

- **Identify the following questions as either background or foreground questions:**

4. **What are the symptoms of acute UTI in infant?**
5. **Can I effectively treat UTI with a shorter course of antibiotics?**
6. **Can this febrile infant be safely treated as an outpatient?**


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Background vs. Foreground

4. **What are the symptoms of acute UTI in infant?**

   Background

5. **Can I effectively treat UTI with a shorter course of antibiotics?**

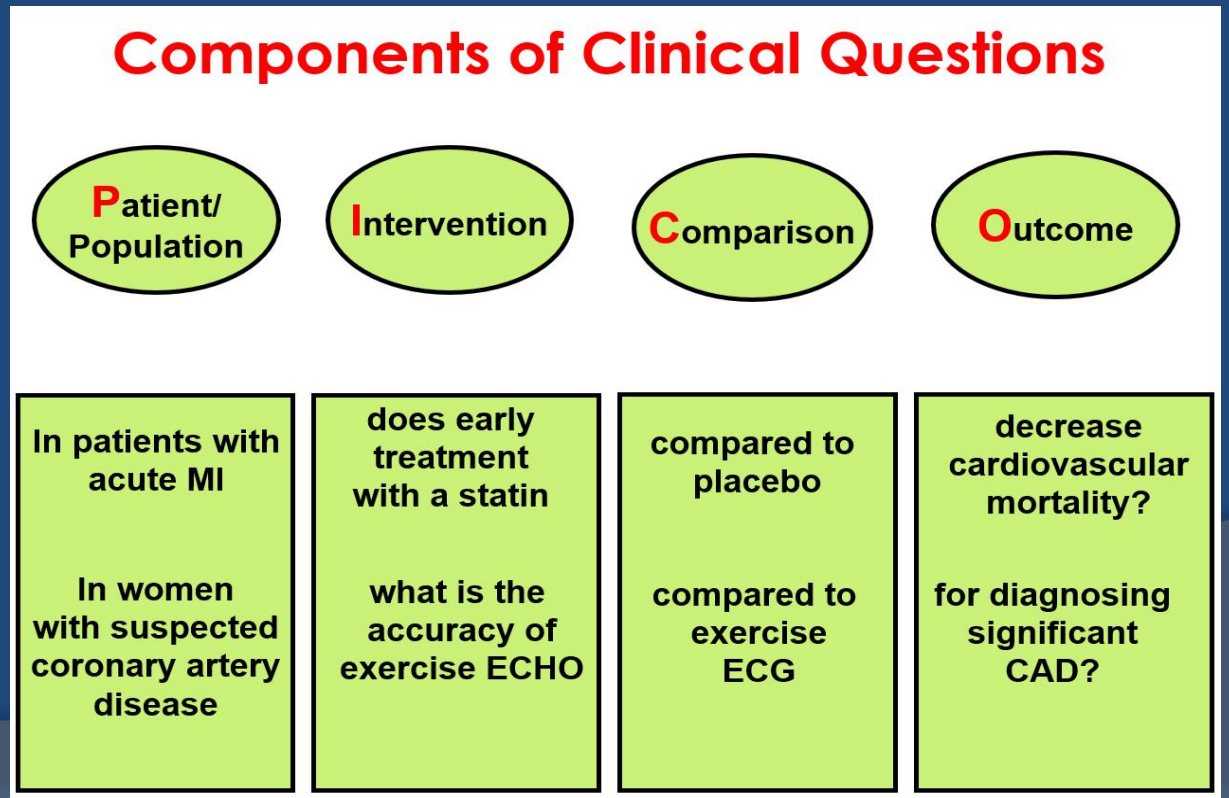
   Foreground

6. **Can this febrile infant be safely treated as an outpatient?**

   Foreground

PICO model

- The PICO model is a tool that can help you formulate a good clinical question.
- Sometimes it's referred to as PICO-T, containing an optional 5th factor.



Anatomy of a Well-Built Question: PICO

- Patient or population – be specific to capture the group you want
- Intervention or exposure – be specific
- Comparison – compare to standard therapy or test
- Outcome – what are the outcomes of interest, be precise

PICO Framework illustrated

Clinical Encounter

Ali, 30 years old, was diagnosed with heart failure at 4 years old and prescribed a beta-blocker which dramatically improved his symptoms. Ali's 5- year-old daughter, Leda, has been recently diagnosed with chronic symptomatic CHF. Ali asks you, whether his daughter should also be prescribed a beta-blocker.

Question: Is there a role for beta-blockers in the management of heart failure in children?

Patient	Children with congestive heart failure
Intervention	Carvedilol (a β-blockers)
Comparison	No carvedilol
Outcome	Improvement of CHF symptoms



PICO Framework illustrated

Clinical Encounter

Mary, 52 years old, was diagnosed with type 2 diabetes 8 years ago. She has been taking metformin, and her blood glucose control improved significantly. Recently, Mary's 45-year-old brother, John, was also diagnosed with type 2 diabetes. John asks whether he should also start taking metformin to help control his blood sugar.

PICO Framework illustrated

Question

Is there a role for metformin in the management of type 2 diabetes in adults?



PICO Content

P — Patient	Adults with type 2 diabetes
I — Intervention	Metformin
C — Comparison	No metformin or lifestyle modification alone
O — Outcome	Improvement in blood glucose control

Practice

Childhood seizures are common and frightening for the parents and the decision to initiate prophylactic treatment after a first fit is a difficult one. To help parents make their decision, you need to explain the risk of further occurrences following a single seizure of unknown cause.



P	***Population/patient***	=	children
I	***Intervention/indicator***	=	one seizure of unknown cause
C	***Comparator/control***	=	no seizures
O	***Outcome***	=	further seizures

Question:

'In children who have had one seizure of unknown cause, compared with children who have had no seizures, what is the increased risk of further seizures?'

Why PICO?

- To get the questions clear in your mind
- To identify the information you need to answer the question
- To translate the question into searchable terms
- To develop and refine your search approach

PICO Components

P - Patient, Population, or Problem	**What are the most important characteristics of the patient? How would you describe a group of patients similar to yours?**
I -Intervention,Exposure, Prognostic Factor	What main intervention, prognostic factor, or exposure are you considering? What do you want to do for the patient (prescribe a drug, order a test, etc.)?
C - Comparison	What is the main alternative to compare with the intervention?
O - Outcome	What do you hope to accomplish, measure, improve, or affect?
T - Time Factor, Type of Study (optional)	How would you categorize this question? What would be the best study design to answer this question?

Types of questions

- Therapy – concerning the effectiveness of a treatment or preventative measure
- Diagnosis – concerning the ability of a test to predict the likelihood of a disease
- Prognosis - concerning outcome of a patient with a particular condition
- Harm - concerning the likelihood of a therapeutic intervention to cause harm

Acquire the Evidence

- Literature Search
- Select a resource
- Consult your local librarian for extra help

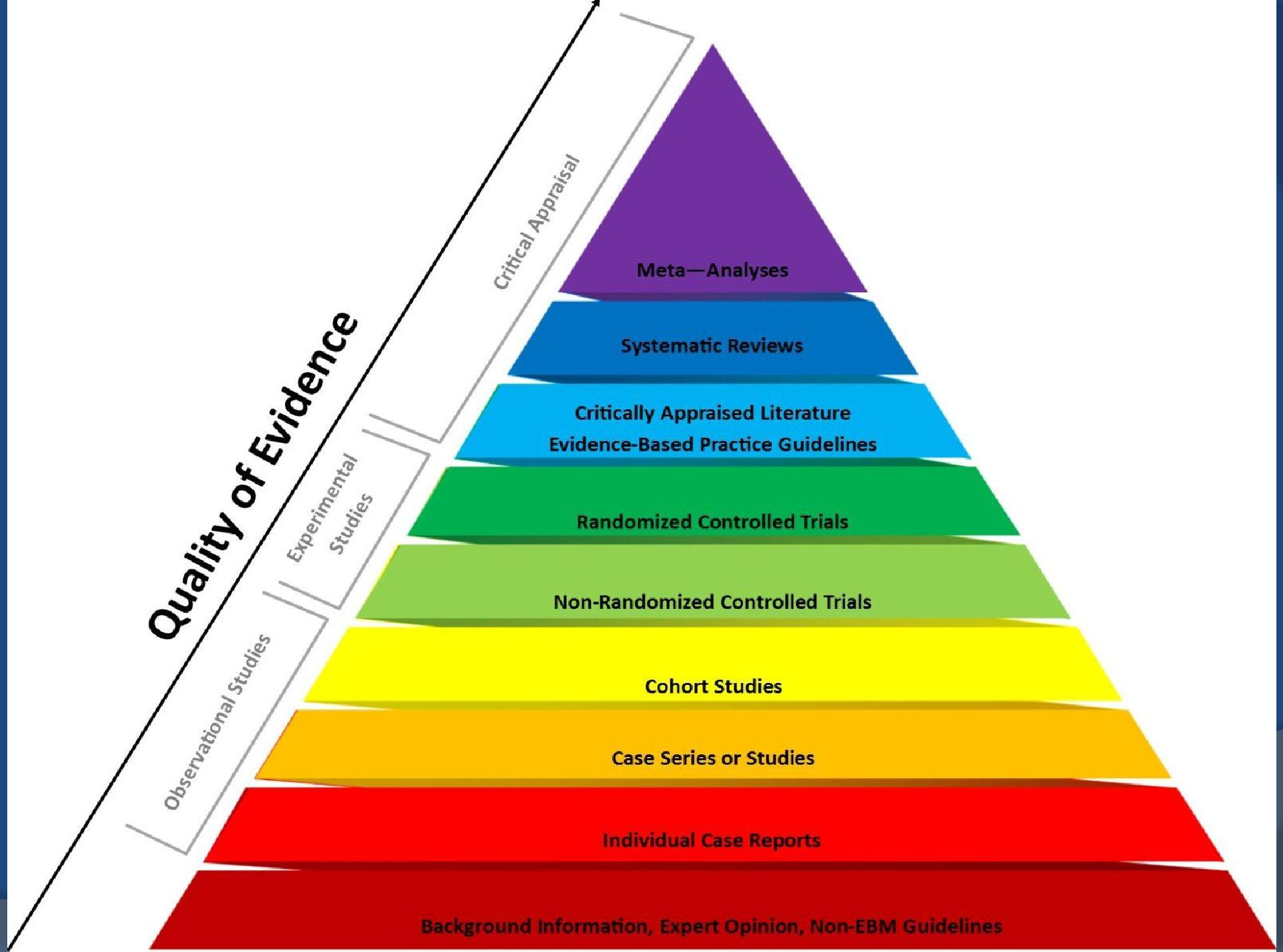
Searching evidence

"My students are dismayed when I say to them, half of what you are taught as medical students today, will have been shown to be wrong in 10 years, and the trouble is, none of us knows which half!"

≈ Dr. Sydney Burwell



- **Primary Literature:** articles and studies presented in peer-reviewed journals.
- **Secondary Literature:** compiled by indexing and abstracting services that can be used to systematically locate various types of published literature through various databases like *Medline (PubMed), Cochrane Library, Ovid, Embase* etc.
- **Tertiary Literature:** core knowledge established via primary literature or accepted as standard of practice within the medical community. The tertiary reference may consist of textbooks/handbooks/manuals on various drugs or disease topics (*Harrison's Principles of Internal Medicine*), compendia (*Physician's Desk Reference*) etc.


Quality of Evidence
Critical Appraisal
Experimental Studies
Observational Studies
Meta—Analyses
Systematic Reviews
Critically Appraised Literature
Evidence-Based Practice Guidelines
Randomized Controlled Trials
Non-Randomized Controlled Trials
Cohort Studies
Case Series or Studies
Individual Case Reports
Background Information, Expert Opinion, Non-EBM Guidelines

TYPE OF QUESTION: WHAT TYPE OF STUDY?

Type of Question	Suggested best type of Study
Therapy	RCT>cohort > case control > case series
Diagnosis	Cohort study >prospective, blind comparison to a gold standard
Etiology/Harm	RCT > cohort > case control > case series
Prognosis	cohort study > case control > case series
Prevention	RCT>cohort study > case control > case series
Cost	economic analysis

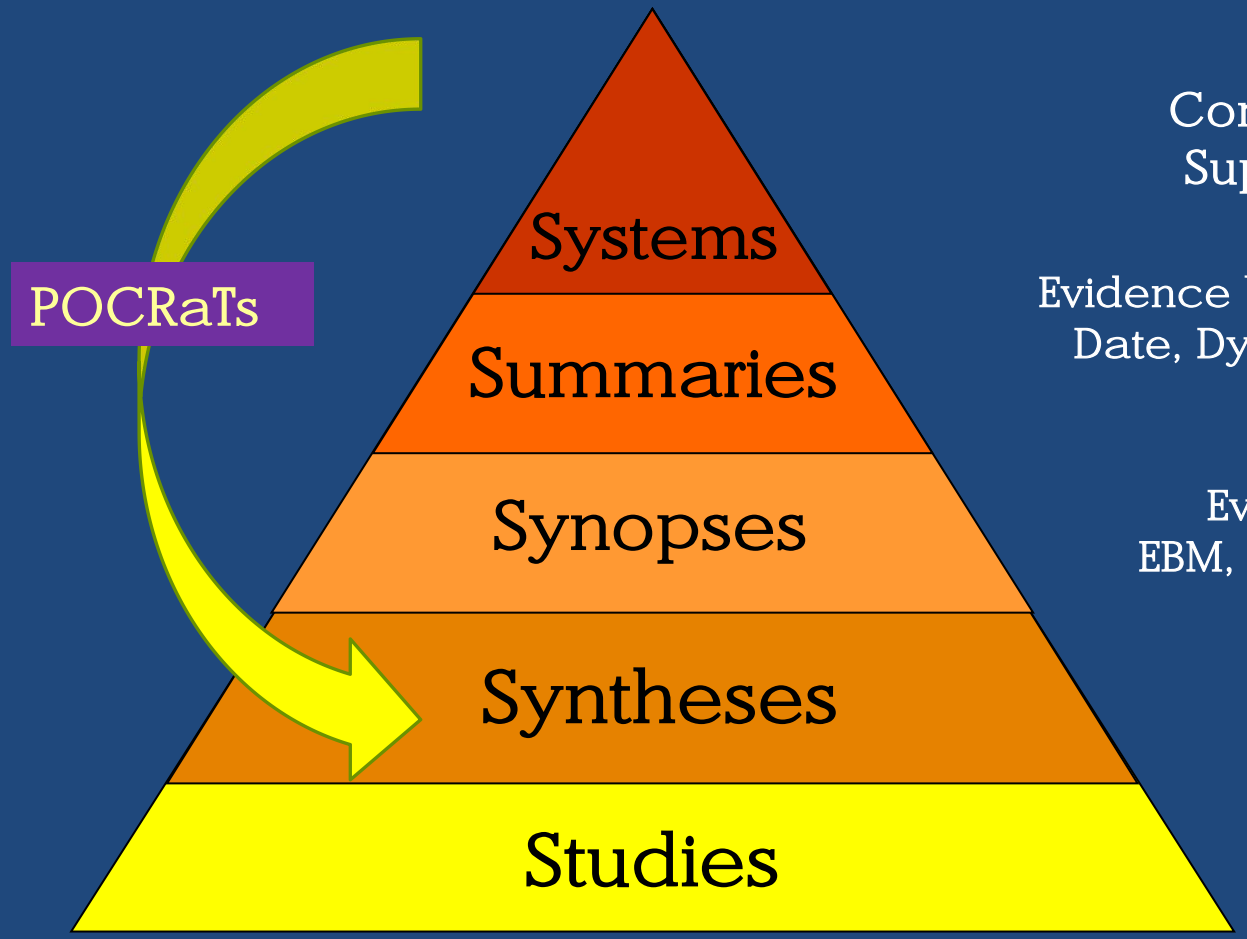
Searching for Research Evidence Based on Research Questions

If your question is about…	Look for a…
Intervention/Therapy	Randomized controlled trial
Diagnosis/Screening To assess accuracy of test To assess effect of test on health outcome	 Cohort study Randomized controlled trial
Prognosis	Cohort study
Etiology/Risk factors/Harm	Randomized controlled trial Cohort study Case-control study

4 Categories of Evidence

- Studies: unfiltered original studies
  - Medline, PubMed
- Summaries: systematic reviews
  - Cochrane
- Synopses: preappraised resource journals
  - ACP Journal
- Systems
  - Clinical Evidence, Up to Date

5’ S’ Information Resources: Information in top 4 are used



Computerized Decision Support System (CDSS)

Evidence based textbooks: UpTo Date, DynaMed, ACP PIER, BMJ Clinical Evidence

Evidence based journals EBM, EBN, EBMH, ACP J club

Systematic Review Cochrane reviews

Original articles: BMJUpdates, PubMed, Clinical Queries

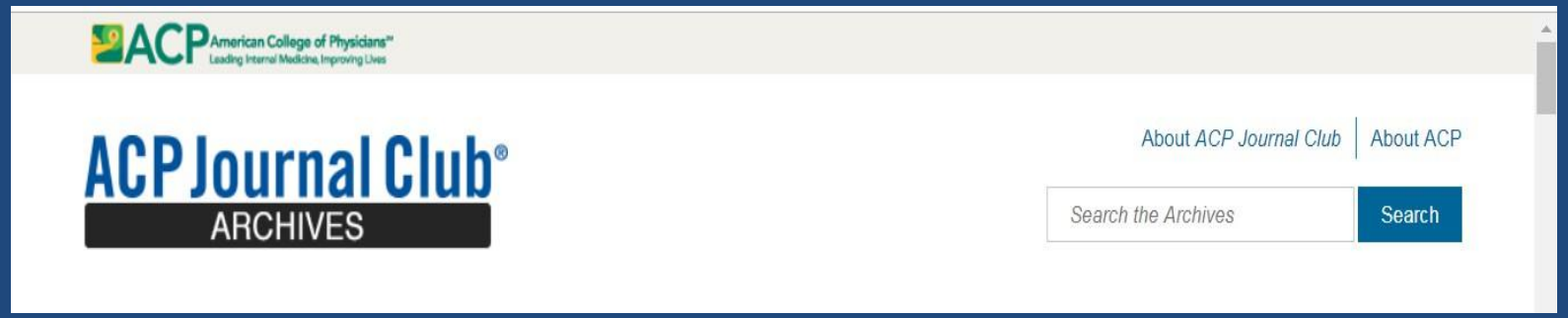
Haynes, R. (2006, November). Of studies, syntheses, synopses, summaries, and systems: the 5S evolution of information services for evidence-based health care decisions. ACP Journal Club, 145(3), A8-A9.

Unfiltered Resources

- PubMed and Medline
  - From peer review journals
  - Good quality articles
  - Use “Clinical Queries” in PubMed
- Google Scholar
  - Grey literature (unpublished or unappraised)
  - Rank in order of most popular cited article
  - Can do “advanced Google search”

Paid Pre-Appraised Resources

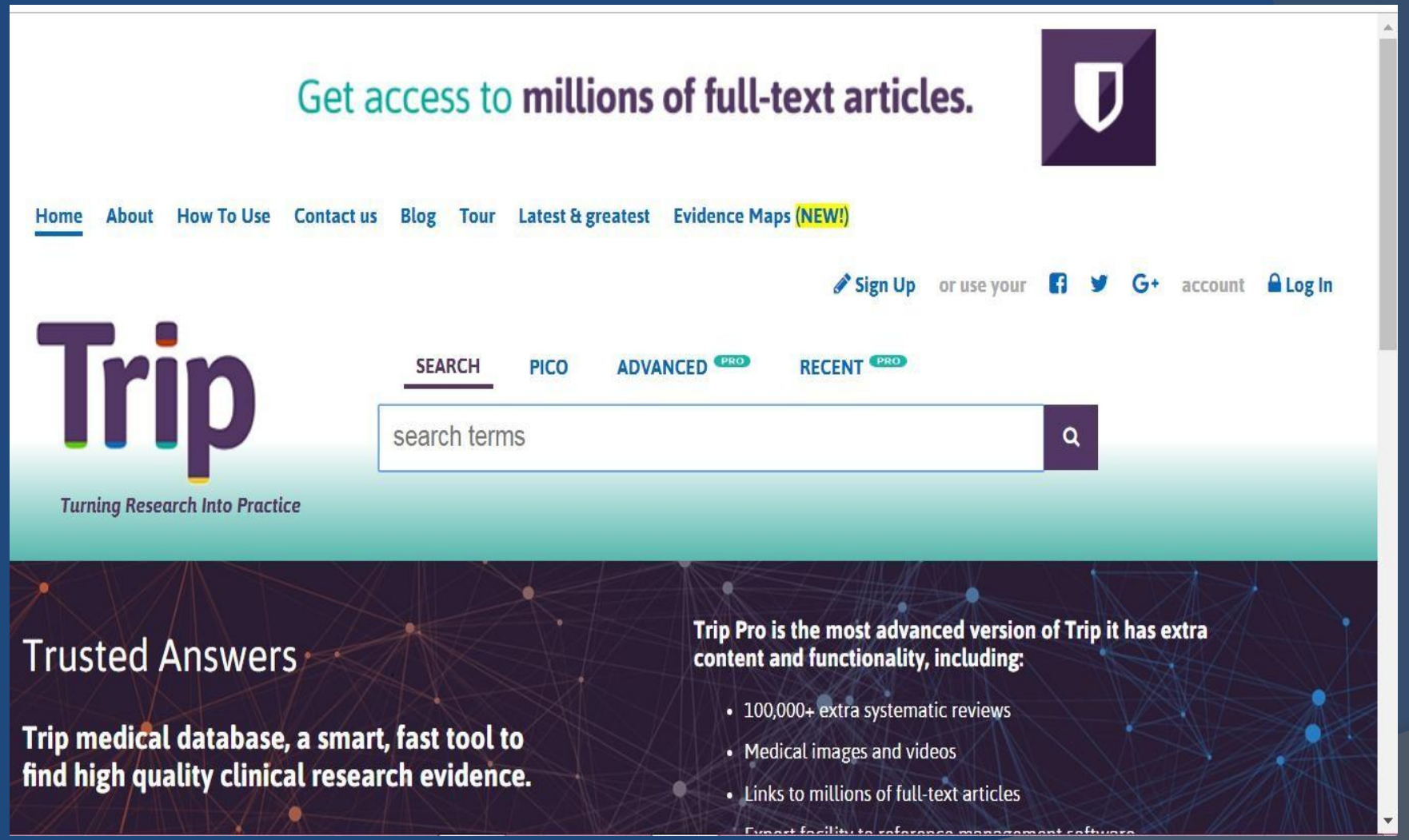
- ACP Journal Club
  - http://www.acpjc.org

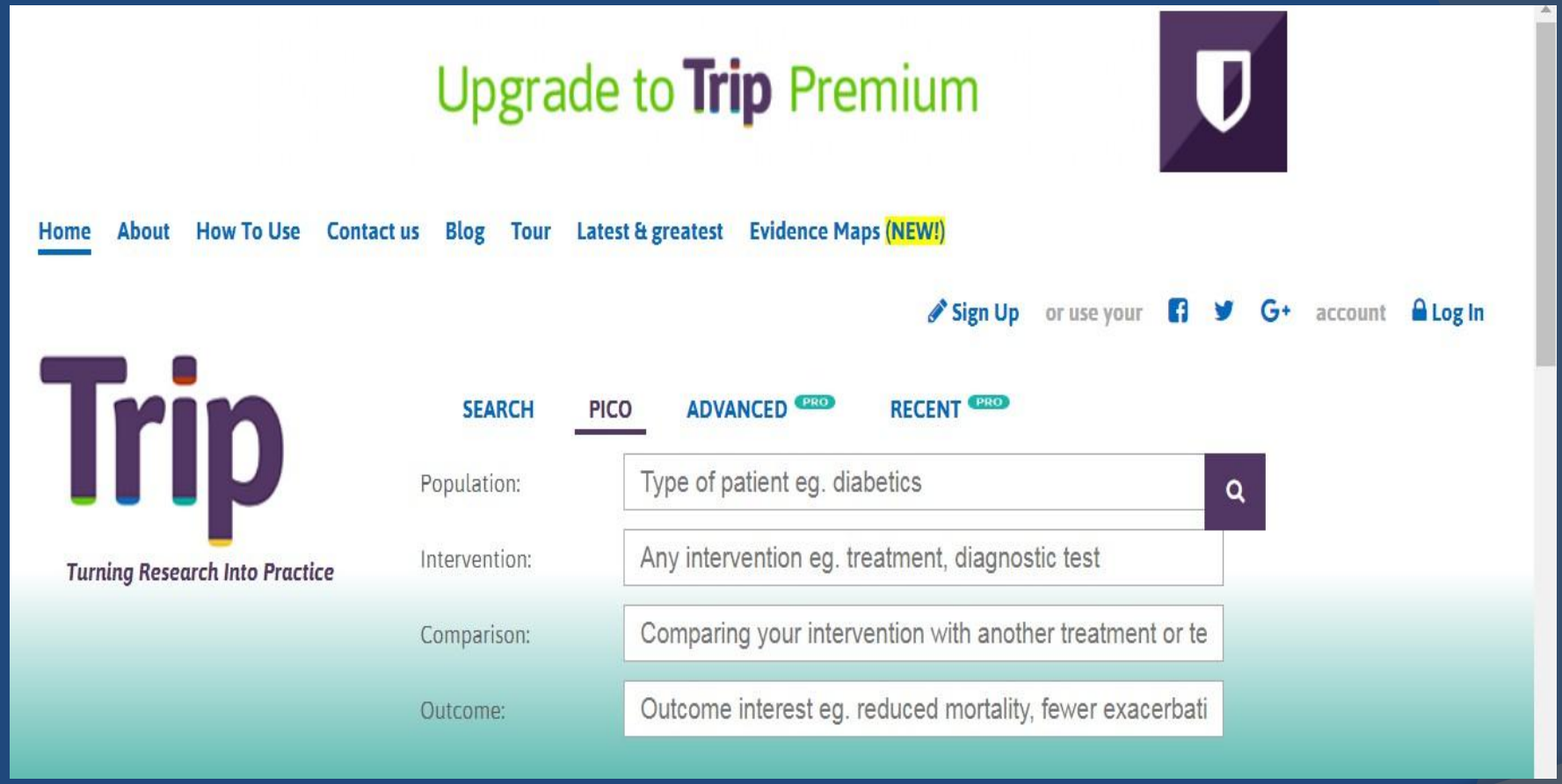


- Clinical Evidence
  - https://www.bmj.com/specialties/clinical-evidence

Free Pre-Appraised Resources

- Cochrane
  - http://www.thecochranelibrary.com
- National Guidelines Clearinghouse
  - https://www.ahrq.gov/gam/index.html
- Best Evidence Topics
  - http://www.bestbets.org
- TRIP Database
  - http://www.tripdatabase.com

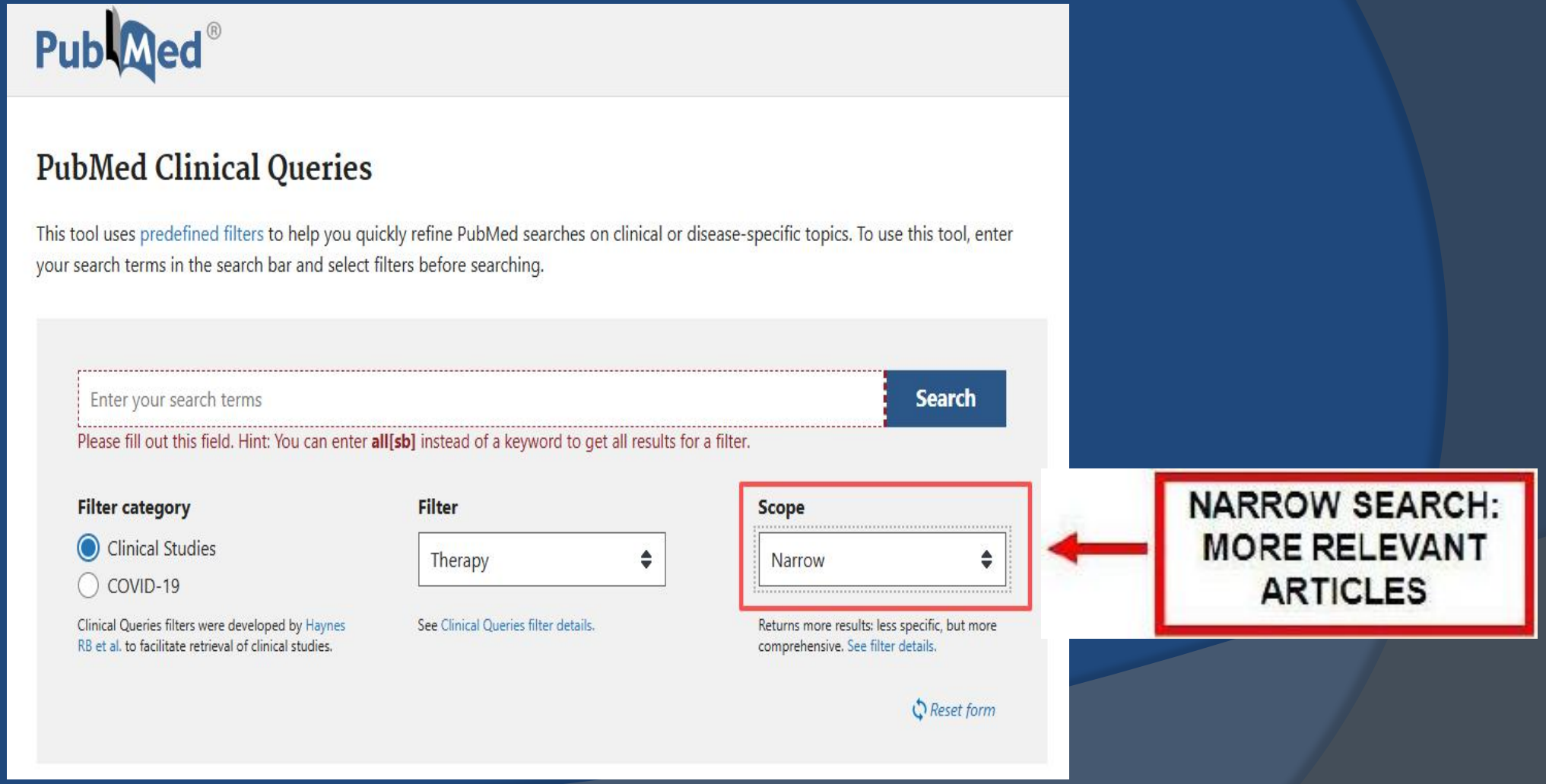

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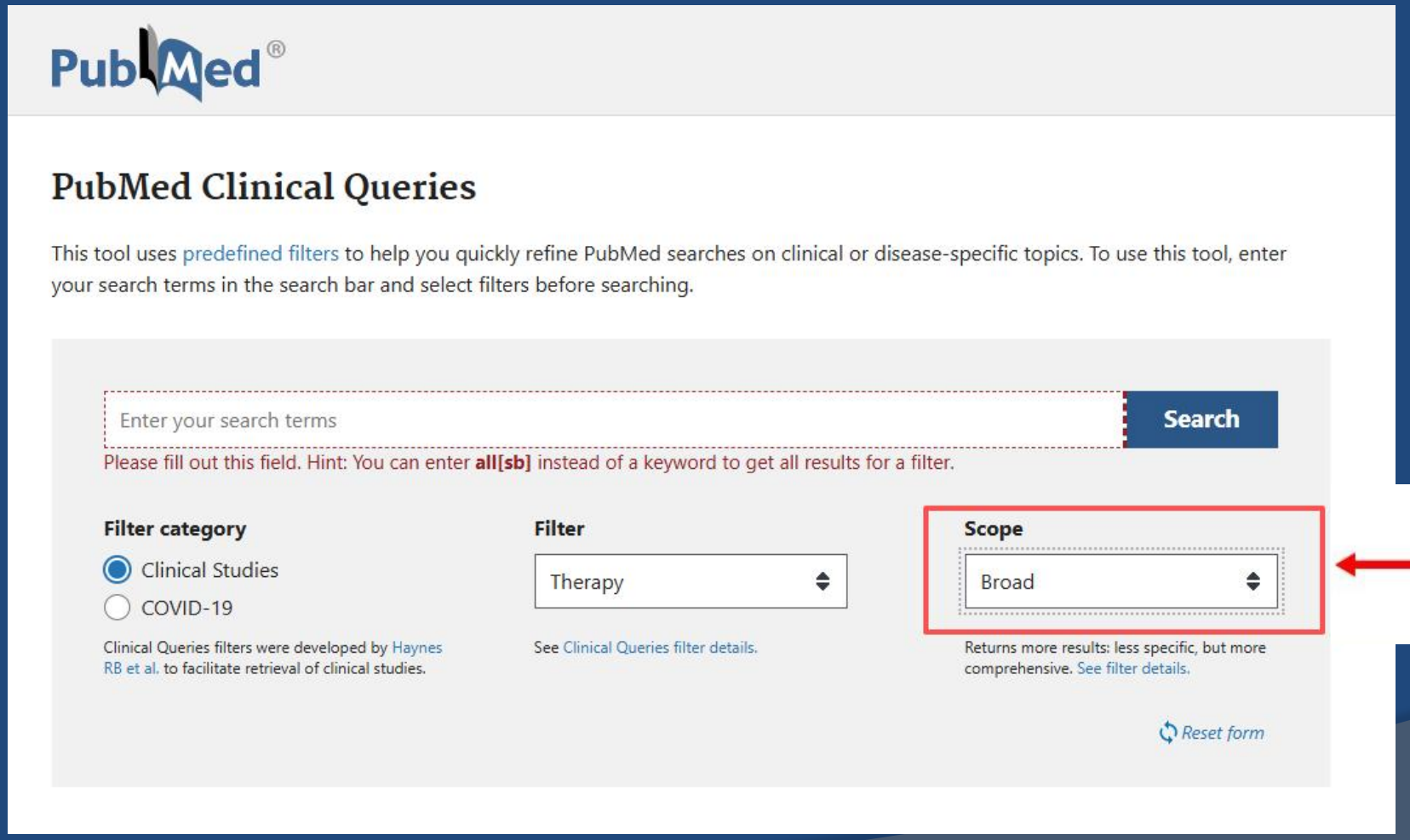
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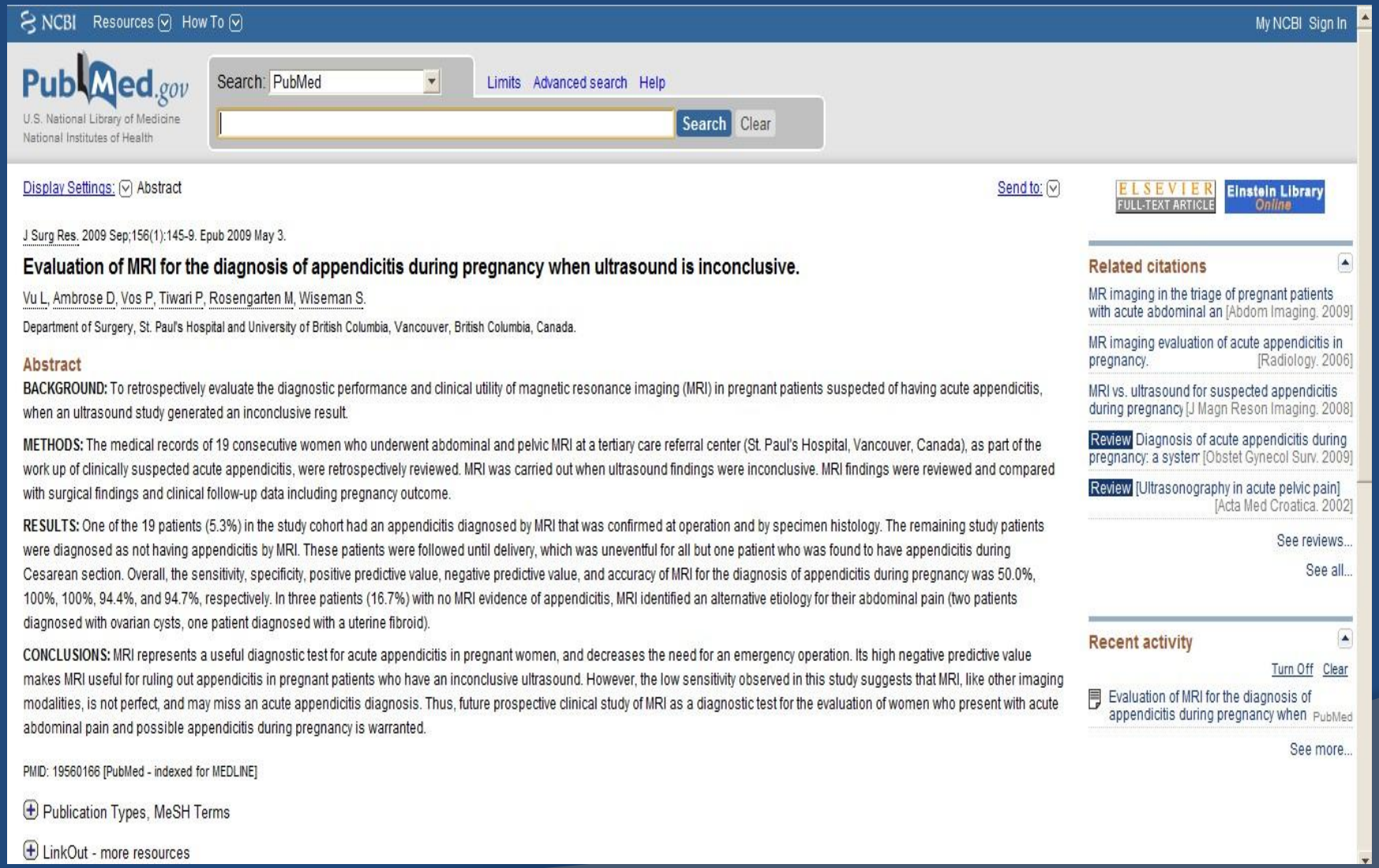
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J Surg Res. 2009 Sep;156(1):145-9. Epub 2009 May 3.

Evaluation of MRI for the diagnosis of appendicitis during pregnancy when ultrasound is inconclusive.

Vu L, Ambrose D, Vos P, Tiwari P, Rosengarten M, Wiseman S.

Department of Surgery, St. Paul's Hospital and University of British Columbia, Vancouver, British Columbia, Canada.

Abstract

BACKGROUND: To retrospectively evaluate the diagnostic performance and clinical utility of magnetic resonance imaging (MRI) in pregnant patients suspected of having acute appendicitis, when an ultrasound study generated an inconclusive result.

METHODS: The medical records of 19 consecutive women who underwent abdominal and pelvic MRI at a tertiary care referral center (St. Paul's Hospital, Vancouver, Canada), as part of the work up of clinically suspected acute appendicitis, were retrospectively reviewed. MRI was carried out when ultrasound findings were inconclusive. MRI findings were reviewed and compared with surgical findings and clinical follow-up data including pregnancy outcome.

RESULTS: One of the 19 patients (5.3%) in the study cohort had an appendicitis diagnosed by MRI that was confirmed at operation and by specimen histology. The remaining study patients were diagnosed as not having appendicitis by MRI. These patients were followed until delivery, which was uneventful for all but one patient who was found to have appendicitis during Cesarean section. Overall, the sensitivity, specificity, positive predictive value, negative predictive value, and accuracy of MRI for the diagnosis of appendicitis during pregnancy was 50.0%, 100%, 100%, 94.4%, and 94.7%, respectively. In three patients (16.7%) with no MRI evidence of appendicitis, MRI identified an alternative etiology for their abdominal pain (two patients diagnosed with ovarian cysts, one patient diagnosed with a uterine fibroid).

CONCLUSIONS: MRI represents a useful diagnostic test for acute appendicitis in pregnant women, and decreases the need for an emergency operation. Its high negative predictive value makes MRI useful for ruling out appendicitis in pregnant patients who have an inconclusive ultrasound. However, the low sensitivity observed in this study suggests that MRI, like other imaging modalities, is not perfect, and may miss an acute appendicitis diagnosis. Thus, future prospective clinical study of MRI as a diagnostic test for the evaluation of women who present with acute abdominal pain and possible appendicitis during pregnancy is warranted.

PMID: 19560166 [PubMed - indexed for MEDLINE]

Publication Types, MeSH Terms

LinkOut - more resources

Related citations

MR imaging in the triage of pregnant patients with acute abdominal an [Abdom Imaging. 2009]

MR imaging evaluation of acute appendicitis in pregnancy. [Radiology. 2006]

MRI vs. ultrasound for suspected appendicitis during pregnancy [J Magn Reson Imaging. 2008]

Review Diagnosis of acute appendicitis during pregnancy: a systerr [Obstet Gynecol Surv. 2009]

Review [Ultrasonography in acute pelvic pain] [Acta Med Croatica. 2002]

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Appraise the Evidence

3 main questions

- Are the results of the study valid?
- What are the results?
- Will the results help in caring for my patient?

Appraise: Therapy

- ARE THE RESULTS VALID?
  - Were patients randomized?
  - Was group allocation concealed?
  - Were groups similar at the start of the trial?
  - To what extent was the study blinded?
  - Was follow-up complete?
  - Were patients analyzed in the groups to which they were first allocated?
  - Aside from the intervention were the groups treated equally?

Appraise: Therapy

- WHAT ARE THE RESULTS?
  - How large was the treatment effect? Relative risk reduction, absolute risk reduction, number needed to treat
  - How precise was the estimate of treatment effect? Confidence interval
  - Were the study patients similar to my population of interest?
  - Were all clinically important outcomes considered?
  - Are the benefits worth the harms and costs?

Appraise: Diagnosis

- ARE THE RESULTS VALID?
  - Was there an independent, blind comparison with a reference standard?
  - Did the patient sample include an appropriate spectrum of the sort of patients to whom the diagnostic test will be applied in clinical practice?
  - Did the investigators perform the same reference standard to all patients regardless of test result?
  - Were the test methods described clearly enough to permit replication?

Appraise: Diagnosis

- WHAT ARE THE RESULTS?
  - Calculate likelihood ratio, estimates the ability of the test to change your pretest probability of disease
  - Will the test be reproducible and well interpreted in my practice setting?
  - Will the test results change my management?
  - Will my patients be better off because of the test?

Appraise: Harm

- ARE THE RESULTS VALID?
  - Were there similar comparison groups with respect to important determinants of outcome other than the one of interest?
  - Were outcomes and exposures measured in the same way in the groups being compared?
  - Was follow up of patients complete?

Appraise: Harm

- WHAT ARE THE RESULTS?
  - Look at Relative Risk or Odds Ratio to estimate the strength of association between the exposure and outcome
  - Is there a dose-response relationship between exposure and outcome?
  - What is the magnitude of the risk?
  - What is the balance between benefits and harms for patients like yours?

Appraise: Prognosis

- ARE THE RESULTS VALID?
  - Was there a representative and well defined sample of patients?
  - Was there a clear description of inclusion and exclusion criteria?
  - Was there adjustment for important prognostic factors?
  - Were objective and unbiased outcome criteria used?

Appraise: Prognosis

- WHAT ARE THE RESULTS?
  - To estimate prognostic risk, look at absolute risk (e.g. 5 year survival rate), relative risk (e.g. risk from a prognostic factor), or cumulative events over time (e.g. survival curves)
  - What are the possible outcomes and how likely are they to occur over time?
  - Will the results lead directly to selecting therapy?
  - Are the results useful for counseling patients?

Critical Appraisal Worksheet

CMU College of Medicine **Critical Appraisal Worksheet** Libraries

Student Name(s):	Group #:
Clinical Scenario:	
P – Patient or Population Describe the most important characteristics of the patient, such as age, disease or condition, gender, etc.	
I – Intervention; Prognostic Factor; Exposure Describe the main intervention, such as drug therapy or other treatment, diagnostic or screening tests, etc.	
C – Comparison Describe the main alternative being considered, such as standard therapy, placebo, or no treatment.	
O – Outcome What is the goal of the patient outcome, such as reduced mortality or morbidity, improved condition and/or quality of life, accurate and timely diagnosis, etc.?	

Check Question Type	Look for Ideal Type of Study
☐ Therapy	Randomized Controlled Trial (RCT)
☐ Prevention	RCT > Cohort Study > Case Control
☐ Diagnosis	Prospective, Blind Controlled Trial Comparison to Other Care Standard
☐ Prognosis	Cohort Study > Case Control > Case Series/Case Report
☐ Etiology/Harm	RCT > Cohort Study > Case Control
☐ Cost analysis	Economic Analysis

Note: Meta-analyses and systematic reviews, when available, often provide the best answers to clinical questions

Database(s) Searched	
Primary Search Terms or Search Strategy *Include a note if the search was saved & where* **Search Terms for *P*** **Search Terms for *I*** **Search Terms for *C*** **Search Terms for *O*** **or the Search Strategy as a whole** (including Boolean operators and nested terms)	
Article Citation (include author(s), article title, journal title, volume/issue/pages, publication date) *Also include PMID, if available*	
Study Type:	
Summary of Evidence (with explanations): Is the study valid, including: ☐ Is it actually the type of study stated? ☐ Was the study conducted properly? ☐ Are there any statistical errors? ☐ Are there any other factors worth noting? What are the clinical results/effects? Do the study results apply to my patient?	
Conclusion:	

Adapted, with permission, from multiple evidence-based medicine worksheets available at the Dartmouth Biomedical Libraries.

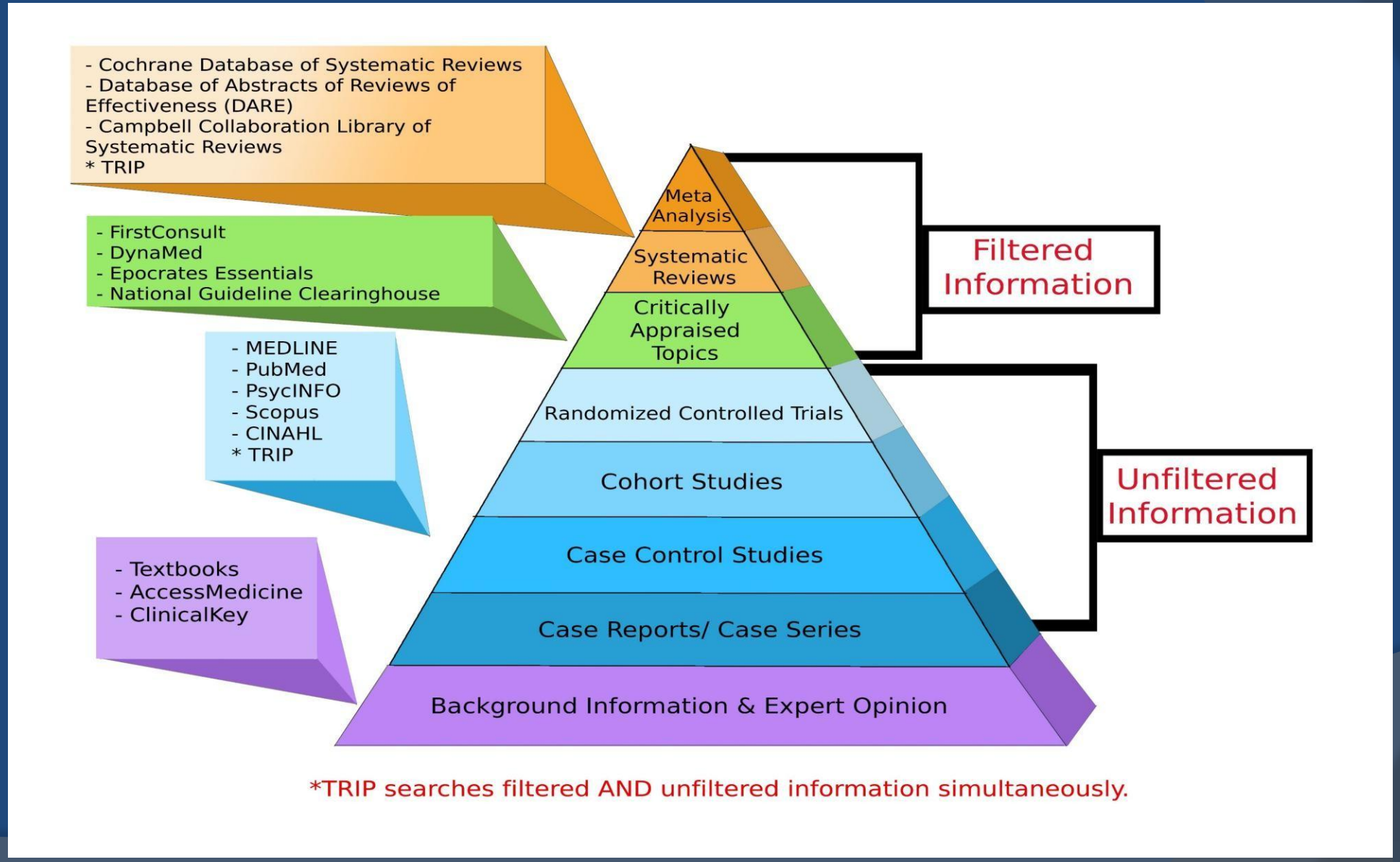
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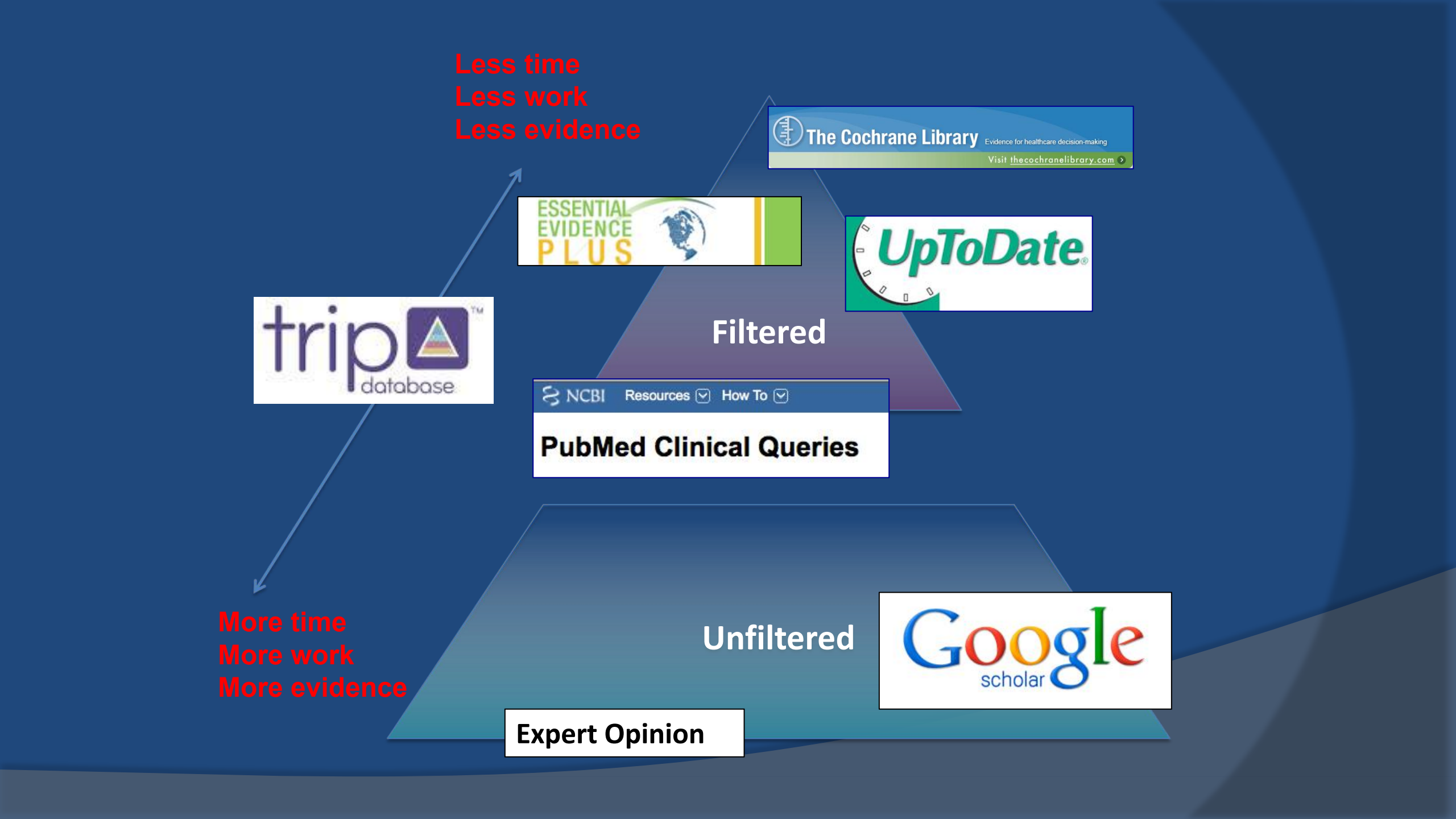
- Reach a conclusion about the answer to the clinical question based on the evidence
- Return to the individual patient
- Combine the evidence and clinical expertise with compassion and patient values

Patient Values

- The unique preferences, concerns and expectations that each patient brings to a clinical encounter and that must be integrated into shared clinical decisions if they are to serve the patient; and by patient circumstances we mean the patient's individual clinical state and the clinical setting

End


- Cochrane Database of Systematic Reviews
- Database of Abstracts of Reviews of Effectiveness (DARE)
- Campbell Collaboration Library of Systematic Reviews
* TRIP
- FirstConsult
- DynaMed
- Epocrates Essentials
- National Guideline Clearinghouse
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- Textbooks
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Outcome – the only thing that matters



What EBM additionally provides is



Opportunity for change
Opportunity for better treatment

Benefits of adopting EBM

- Minimizes the error and optimizes the quality in patient care
- Reduces the cost of treatment
- Helps in advancement of knowledge and keeping pace with scientific progress

Challenges in adopting EBM

- **Technology and online information resources must be *available to the clinicians*.**
- **Understanding of the epidemiological study designs and *concepts of biostatistics* should be clear.**
- ***Attitude of the clinician*- one must realize that clinical performance depends upon regularly updating knowledge and not merely on practical clinical experience.**

Conclusion: What EBM is

NOT	But it is
• "Cookbook" medicine • Rigid adherence to clinical guidelines	• A rigorously systematic way to evaluate the strength and appropriateness of available evidences for a particular clinical situation
• Managed care • Cost-cutting measures	• A way to avoid waste by considering both the efficacy and effectiveness and cost of a particular intervention in a particular clinical setting.

Conclusion: What EBM is

NOT	But it is
• The same thing as clinical epidemiology or biostatistics	• Build on concepts so we can better understand the strength of inferences from available evidence.
• Limited to RCTs	• A recognition that some study designs (esp. RCTs) are less susceptible to bias than others, and therefore less likely to mislead, but other evidences should also be used in clinical decision making as long as we understand their limitations.

Level of Evidence



- Systematic review and Meta-analysis have the **highest level of evidence in** evidence-based medicine

Why meta-analysis?

- Increase the power
- To establish whether there is an effect
- To understand the size of the effect and the certainty around it
- To investigate if the effect is consistent across studies
- To settle controversies arising from apparently conflicting studies or to generate new hypotheses

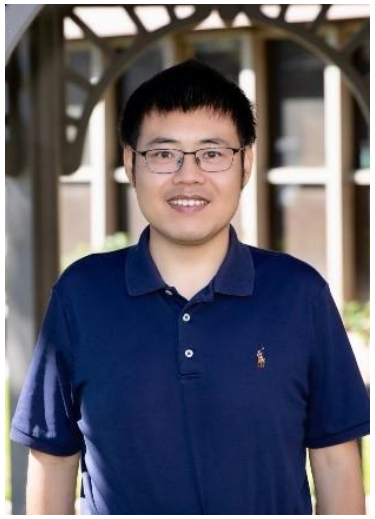

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